

FTTH and Passive Optical Networks

Depending on the termination point, fibre access networks have different configurations, such as fibre-to-the-home, fibre-to-the-premise, fibre-to-the-building, fibre-to-the-curb, or fibre-to-the-node. Let us have a look

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Since long back, telecommunication service providers have dreamed of an all-fibre network and for good reasons. A fibre provides substantially more bandwidth, carries signals farther, is more reliable and secure, and has a longer life span. Additionally, providers view fibre's bandwidth capacity as a competitive weapon, particularly in the access network.

Fibre is seen as the pre-eminent long-term alternative to today's broadband access technologies. One that not only allows providers to generate new services but also provides them with significant and sustainable reductions in operating expenses, and shifts their capital spend-

ing from older technologies to newer, less costly technologies. The single greatest driver for fibre in the access network is multi-play services, the opportunity to offer subscribers high-speed data, voice and video as one of a variety of potential bundled services.

Today, fibre networks come in many varieties, and telecommunication service providers consider a variety of FTTx architectures (here 'x' stands for the termination point) for delivering fibre to their subscribers. Depending on the termination point, FTTx is classified as fibre-to-the-premise (FTTP), fibre-to-the-home (FTTH), fibre-to-the-building (FTTB), fibre-to-the-curb (FTTC) or fibre-to-the-node (FTTN), each having different configurations and characteristics.

Currently, there is no one-size-solves-all architecture, so providers must make a series of technology decisions based on their service goals. A primary consideration for providers is to decide whether to deploy an active (point-to-point) or passive (point-to-multipoint) fibre network.

Configurations

FTTx configurations depend on how the optical fibre is used in the last-mile access network.

FTTH. It is a method of installing optical fibre cable to the home. FTTH is the final configuration of access networks using optical fibre cable. It consists of a single optical fibre cable from the telecom service provider's central office, or base station, to the subscriber's home. An OLT optical line terminal (OLT) at central office connects optical network unit (ONU) residing at subscriber's home through a dedicated fibre. The optical-to-electrical signal conversion is done through a modem and connection to the user's computer is established via an Ethernet card (Fig. 1).

FTTB. In this configuration, optical fibre cable is run from the telecom service provider's central office (OLT/central terminal) or base station to the building's termination point (ONU/remote terminal). A local area network (LAN), or existing telephone metallic cable, is then used to connect the user (Fig. 2).

FTTC. It is a method of installing optical fibre cable by the curb near the user's home. An optical communications system is then used for the remote (optical or electrical) signal conversion unit installed outside (near the curb or on a telephone pole) the installation centre. Finally, metallic

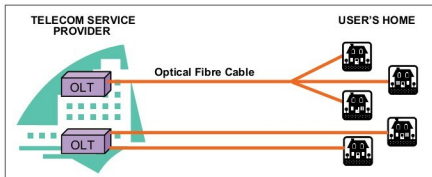


Fig. 1: FTTH configuration

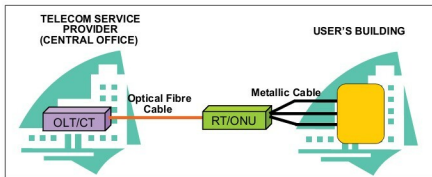


Fig. 2: FTTB configuration